Cart lab: change mass

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a controlled comparison to describe the effect of system mass on acceleration.

SUCCESS CRITERIA - I can...

1. Identify total moving mass and keep driving force fixed.
2. Compare acceleration values using a fair test.
3. Separate supplied or simulated values from values you measured.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A car starts at **+24 m/s** and brakes at **-4 m/s²** until rest. A student extends that acceleration to **8 s**. **Find the stop time**/distance and **explain why extension** is invalid for ordinary braking.

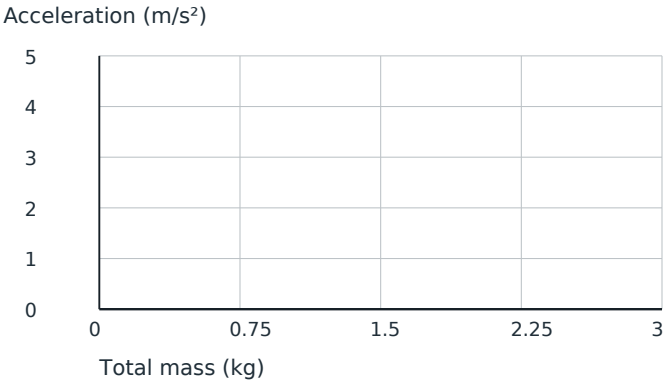
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $a = F_{net}/M = m_h\, g/M$.
Ideal cart + hanger: M is all moving mass; m_h is hanging mass. Hold m_h fixed; neglect resistance and pulley inertia.

Worked reasoning: Supplied cart trials use the same net force: masses **1** and **2 kg** accelerate at **4** and **2 m/s²**. In both cases ma=**4 N**; doubling mass halves acceleration.

Your turn: Plot acceleration against mass for the two trials. **Label what** was held constant.



Marzano check - circle: ① ② ③

03 Test an inverse relationship

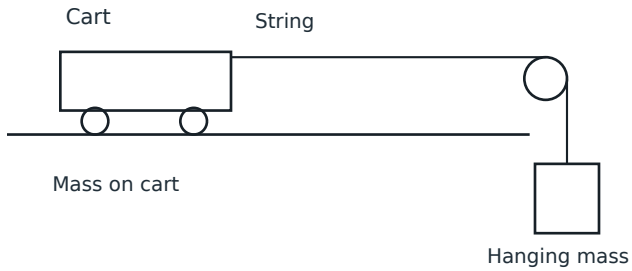
The table contains ideal data for a constant **0.49 N** net force.
1. Calculate **1**/M for each mass.
2. **Sketch acceleration versus 1**/M, then **state the slope's units** and physical meaning.

M (kg)	a (m/s²)
0.50	0.980
0.75	0.653
1.00	0.490
1.50	0.327

Marzano check - circle: ① ② ③

04 System boundary

The cart and hanger accelerate together.
1. Use the setup diagram to **explain why both belong** in the total moving mass and why string tension is internal to the combined system.
2. **State two idealizations**.



Marzano check - circle: ① ② ③

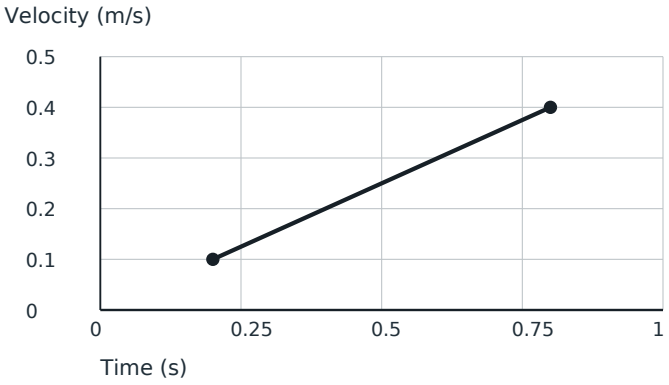
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Continue down each column.

05 Measured slope and repetitions

A sensor records velocity changing from **0.10 m/s** at **0.20 s** to **0.40 m/s** at **0.80 s**. **Find the acceleration** from the graph. Then design a table for three total masses and three trials per mass; **label any data you actually** measure.



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **3 kg** object hangs still from a vertical string. Use g=**9.8 N/kg**. **Name all forces** on it with sources, sizes and directions. Does object-on-string belong on this diagram? **Draw the forces** on the chosen object and **label their sources**.

Add and label force arrows.

Marzano check - circle: ① ② ③

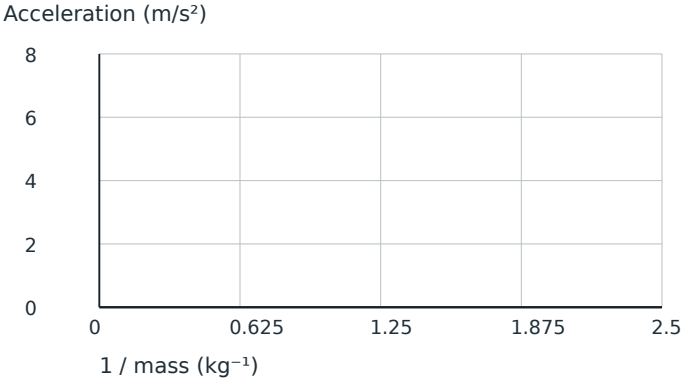
07 Exit check

A **0.80 kg** moving system accelerates at **0.60 m/s²**. At the same net force, predict a for **1.20 kg**. **Explain how added resistance would** change the prediction.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

The cart's acceleration follows a=F/m. At constant net force **3 N**, find accelerations for masses **0.5**, **1**, **2 kg**. **Choose a transformed horizontal axis** that makes a straight-line graph and **state its slope**.



Marzano check - circle: ① ② ③